

Patrick Carroll

Model and Simulation Engineer

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Security Clearance

Active Top-Secret SCI

2014 - Present

Summary

Model & Simulation Engineer specializing in AFSIM plugin development, simulation model integration, and CI/CD automation for defense applications. Combines 10+ years of technical and military experience across software development, engineering automation, systems administration, and RF signal analysis.

Skills

Programming Languages:  C/C++,  Python,  MATLAB,  HTML,  CSS,  JavaScript

Tools and Frameworks: TortoiseSVN, Git, Jira, Confluence, Jenkins, Groovy

Experience

Junior Model and Simulation Engineer	Radiance Technologies, Beavercreek, Ohio	08/2024 - Present
- Develop and maintain CI/CD pipelines to build, test, and validate Simulink models into AFSIM plugins. - Package models and plugins to deploy across multiple government programs and agencies. - Partner with cross-functional engineering teams and government customers to integrate new models and features.		
Information Systems Technician	United States Navy Reserve, Cincinnati, Ohio	04/2019 - 04/2025
Supported the Navy part-time. Trained in Cisco Routers, Switches, End Devices, IPv4 and IPv6 addressing, network protocols, and troubleshooting. Qualified in Enlisted Information Warfare.		
System Software Support Engineer	MBS Professional, Mason, Ohio	01/2024 - 08/2024
- Developed Windows Preinstall scripts for Image Capture/Deployment. - Developed a VBA PowerPoint add-in that automated AI-driven text-to-speech video generation. - Developed VBA macros in MS Word for SQL database interactions.		
IT Systems Administration Assistant	Texas A&M Corpus Christi, Corpus Christi, Texas	08/2019 - 12/2021
Handled 700+ tickets dealing with the administration of User Accounts, E-mail, Listserv, Security Groups, Distribution Groups, Shared Mailboxes, File Systems, HPC and Active Directory.		
Submarine Electronics Technician	United States Navy, Norfolk, Virginia	04/2014 - 04/2019
- Analyzed and classified different types of RF signals. - Led teams of 2-3 in the maintenance and repair of critical systems. - Supported joint operations at sea.		

Projects

Stock Bot, Automated day-trading engine that scans for volatile stocks, analyzes news using GPT-4o, and executes trades via the Interactive Brokers API. Built with Python, and hosted on AWS.

Document AI Organization & Scanner, Document classification, renaming, meta-tagging, and organization using AI. Utilized Microsoft Azure AI Intelligence, OpenAI, Python, Java, and JavaScript.

Battlezone, Recreation of the 1980s tank simulator in C/C++ OpenGL, with iconic Green Vector 3D graphics, particle effects and collision detection.

Comparing LLM Context Similarity, Tested the similarities and accuracy of different OpenAI models with Jupyter Notebooks.

Education

B.S. in Computer Science (GPA 3.65)	Texas A&M-Corpus Christi	04/2019 - 12/2023
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Certifications

CompTIA Security+	06/2022 - Present
Microsoft Certified: Azure AI Fundamentals	07/2023 - Present